

AI Scam Detector

Complete Beginner Roadmap — Portfolio Edition

Your Burning Questions — Answered

■ Frontend first or backend first?

Backend first — always. Build the brain before the face. Without a working classifier, your UI has nothing to show. Start with Python logic, get it working in a plain script, then wrap it in Streamlit.

■ What technologies do I actually need?

Python is all you need to start. Just 4 libraries: pandas (data), scikit-learn (ML), nltk (text processing), Streamlit (UI). Don't install anything else until you need it.

■ Do I need to know ML deeply?

No. You need to understand what a classifier does, not the math behind it. Copy-paste is fine to start — understanding comes from breaking things and fixing them.

■ Do I need a database?

Not for now. CSV files are your database. Keep it simple until the core works.

■ Should I use HuggingFace/transformers?

Only in Phase 3 (optional). Start with scikit-learn — it's simpler, faster to learn, and more than enough for a beginner portfolio project.

Your Tech Stack — Simple & Beginner-Friendly

■ Python

Your only language. Everything runs in Python.

■ pandas

Load & manage your dataset (CSV).

■ scikit-learn

Train your spam classifier model.

■ nltk

Clean & process raw text.

■ Streamlit

Build your UI in pure Python.

■ Install everything in one command:

```
pip install pandas scikit-learn nltk streamlit
```

Week-by-Week Build Plan (3–4 Weeks)

[Week 1] Setup + Data

◆ Install Python, VS Code, Git ◆ Download Kaggle SMS Spam dataset (free, just google it) ◆ Load CSV with pandas, explore it — print first 10 rows ◆ Understand what ham vs spam looks like in the data ◆ Goal: You can load and explore the dataset

[Week 2] Build the Classifier

◆ Clean text: lowercase, remove punctuation (use nltk) ◆ Convert text to numbers using TF-IDF (scikit-learn)
◆ Train Logistic Regression — literally 2 lines of code ◆ Test it: type a message, get spam/not spam output ◆
Goal: Working classifier in a .py script

[Week 3] Add Psychological Detection

◆ Create keyword lists for: Urgency, Fear, Greed, Authority ◆ Write a function that scans message for these keywords ◆ Tag the message with which manipulation type was found ◆ Combine with classifier output ◆
Goal: Message gets score + manipulation label

[Week 4] Build Streamlit UI + Polish

◆ Create app.py — your Streamlit frontend ◆ Text input box + Analyze button ◆ Show: Scam probability bar, manipulation type badge, highlighted risky words ◆ Test with 10+ real scam examples ◆ Deploy free on Streamlit Cloud, write GitHub README

Project Folder Structure

```
scam-detector/ ■■■ data/ ← your CSV dataset goes here ■■■ model/ ← saved trained model  
■■■ train.py ← train your classifier ■■■ detector.py ← core logic (detection +  
patterns) ■■■ app.py ← Streamlit UI ■■■ requirements.txt ← list of libraries ■■■  
README.md ← explain your project (important!)
```

Do's and Don'ts for Beginners

DO

- Start with small working code — don't over-plan
- Break code into functions (train.py, detector.py, app.py)
- Google every error — this is normal, even for seniors
- Commit to GitHub every time something works
- Test with real scam messages you find online

DON'T

- Don't try to build everything in one file
- Don't spend days on model accuracy — ship first
- Don't skip the README — it's half your portfolio
- Don't install 20 libraries you don't understand yet
- Don't aim for perfect — aim for working

Free Resources to Learn As You Build

Dataset	Kaggle SMS Spam Collection	kaggle.com → search 'SMS Spam Collection'
Python Basics	Python.org Tutorial	docs.python.org/3/tutorial
scikit-learn	Official Quickstart	scikit-learn.org/stable/getting_started
Streamlit	30 Days of Streamlit	30days.streamlit.app
Deploy Free	Streamlit Community Cloud	streamlit.io/cloud
Version Control	Git & GitHub Basics	github.com → search 'Git for beginners'

■ Remember: A working, ugly project beats a perfect project that doesn't exist.